

PGRPLB (Anopheles gambiae, UniProt F5HMW5) — Hypothesis Deep Research

Focus: function_assignment · **Hypothesis slug:** immune-receptor-defense-and-secretion
Source: genes/ANOGA/PGRPLB/PGRPLB-ai-review.yaml (free-text)

Seed hypothesis (restated)

F5HMW5 (1) has peptidoglycan **immune receptor activity**, (2) contributes to **defense against Gram-positive bacteria**, and/or (3) **localizes extracellularly** — each to be evaluated *separately from* amidase activity.

Each sub-claim maps precisely onto an existing **IBA-only** (phylogenetically inferred) GO annotation on F5HMW5: (1) → **GO:0016019** peptidoglycan receptor activity (MF, IBA); (2) → **GO:0050830** defense response to Gram-positive bacterium (BP, IBA); (3) → **GO:0005576** extracellular region (CC, IBA).

Executive Judgment

- **Sub-claim 1 — peptidoglycan immune *receptor* activity (GO:0016019): OVER-ANNOTATED / refuted as a core function.** F5HMW5 is a bona fide *catalytic* amidase of the PGRP-LB subfamily (PANTHER SF77), with a fully intact zinc/catalytic-Cys active site that aligns residue-for-residue with the biochemically characterized *Drosophila* PGRP-LB amidase. In this pathway the sensing receptor is **PGRP-LC** (*Anopheles* PGRP-LC), not PGRP-LB. The GO:0016019 IBA is propagated from a mixed PANTHER node containing recognition PGRPs/mammalian PGLYRPs — a classic paralog/family over-annotation. PGRP-LB binds peptidoglycan as a **substrate for catalysis**, not for signal transduction.
- **Sub-claim 2 — defense against Gram-positive bacteria (GO:0050830): weakly supported / mis-directed.** PGRP-LB is a **negative regulator** of the Imd pathway (dampens immunity), and its amidase specifically degrades **DAP-type (Gram-negative)**

peptidoglycan. In *Anopheles*, PGRPLB *promotes* permissiveness to *P. falciparum* — i.e., it suppresses rather than mediates defense. A "defense response to Gram-positive bacterium" term is questionable on both chemotype (DAP vs Lys) and sign (suppressor vs effector).

- **Sub-claim 3 — extracellular localization (GO:0005576): partially supported (orthology), unresolved for this accession.** The *Drosophila* ortholog is experimentally secreted, but F5HMW5 carries **both "Signal" and "Transmembrane" keywords** with a predicted N-terminal helix (7–23); the N-terminus could be a cleaved signal peptide (secreted) or an uncleaved signal anchor (type-II membrane). Crucially for curation: F5HMW5 = AGAP001212 with a **single annotated transcript** (AGAP001212-RA), **protein existence PE3 (inferred from homology)**, and the transmembrane call is a **prediction-only (SAM automatic)** annotation with no experimental support. The multi-isoform localization argument from *Drosophila* therefore does not transfer, and localization is best treated as an orthology-based lead rather than an established fact.
- **Anopheles-specific receptor architecture (Iteration 3):** In *Anopheles*, the peptidoglycan **receptor** of the Imd pathway is PGRPLC (transmembrane; binds DAP-type PGN by pull-down;

),
P 31760136
 with PGRP-S1 serving Toll
 (
P 39761205
).

This species-matched evidence assigns "peptidoglycan receptor activity" to PGRPLC, **not** PGRPLB — directly reinforcing that GO:0016019 on F5HMW5 is paralog-borrowed over-annotation.

Overall verdict: the seed hypothesis is largely NOT supported as stated. The gene product's primary, best-supported function is a **secreted/extracellular DAP-type peptidoglycan amidase that negatively regulates the Imd immune response** — not a signaling receptor and not a Gram-positive defense effector.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
UniProt F5HMW5 (this run)	Computational (domain/ sequence)	Refutes receptor; supports amidase	Is F5HMW5 a catalytic amidase or a receptor?	Pfam Amidase_2 (PF01510); PANTHER PGRP-LB (SF77); intact catalytic Cys (ECPG, C178) + zinc-His (GAH/GHR); near-identical to Dros PGRP- LB catalytic core	A. gambiae protein, 205 aa	High for catalytic assignment in-silico active site (no direct enzyme assay on F5HMW5)
 16618604 (Zaidman- Rémy 2006)	Direct assay + localization	Refutes Gram- positive & receptor; qualifies secretion	Ortholog function/ localization/ specificity	"PGRP-LB is a secreted protein"; "amidase that specifically degrades gram- negative bacteria peptidoglycan"; "downregulates the Imd pathway"	Drosophila	High; ortholog (n A. gambiae)
 28494453 (Gendrin 2017)	Mutant/RNAi phenotype	Refutes defense- effector & receptor	In-vivo role of Anopheles PGRPLB	PGRPLC is the receptor activating Imd; PGRPLB negatively regulates Imd; "PGRPLB promotes mosquito permissiveness	Anopheles coluzzii/ gambiae, gut/ systemic	High; species- matched primary evidence

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
				to <i>P. falciparum</i> "		
P 22118526 (Paredes 2011)	Mutant phenotype	Supports negative-regulator BP	Is PGRP-LB a negative regulator?	Confirms "PGRP-LB as a negative regulator of the Imd pathway"; catalytic PGRPs scavenge peptidoglycan	<i>Drosophila</i>	High; ortholog
P 31760136 (Rodgers/Gendrin 2020)	Interaction + phenotype	Refutes receptor claim for LB	Which <i>Anopheles</i> PGRP is the PGN receptor?	PGRPLC is "the transmembrane receptor of the Imd pathway"; PGRPLC isoforms pulled down with DAP-type peptidoglycan (LC1/LC3 polymeric, LC2/LC3 TCT)	<i>Anopheles coluzzii</i> midgut	High; species-matched; concerns L (assigns receptor ro away from LB)
P 39761205 (Song 2025)	Mutant phenotype	Qualifies (chemotype/receptors)	Which receptors sense PGN in mosquito gut?	Mosquito PGN receptors are Pgrp-lc (IMD) and Pgrp-s1 (Toll); both DAP-PGN and Lys-PGN present in gut	<i>Anopheles stephensi</i> gut	Medium-High; supports receptor=L and chemotype nuance
UniProt xref (this run, Iter 3)	Database	Qualifies localization	Gene identity, isoforms, evidence	F5HMW5 = VectorBase/Ensembl AGAP001212	<i>A. gambiae</i>	High for identity; topology

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
			level of F5HMW5	(single transcript AGAP001212- RA); protein existence PE3 (inferred from homology) ; N- terminal TM 7-23 is SAM automatic prediction only		prediction- only
Comparative alignment (this run, Iter 2)	Structural/ evolutionary (computational)	Refutes receptor; supports amidase	Does F5HMW5 keep catalytic vs recognition features?	F5HMW5 most identical to catalytic amidases (PGLYRP2 57.8%, PGRP-LB 56.7% > recognition LC 47.8%, SA 36.7%); diagnostic catalytic Cys (ECPG/C178) conserved with PGRP-LB & PGLYRP2, diverged in recognition PGRPs	A. gambiae protein vs PGRP panel	Medium- High; NW identity over aligned columns is approxima SC1 anchor caveat note
QuickGO annotations F5HMW5 (this run)	Database	Context	What is currently annotated?	GO:0016019, GO:0050830, GO:0005576 are IBA-only ; amidase	—	Database orientation IBA nodes mix recognition

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
				(GO:0008745), Zn (GO:0008270), PGN binding (GO:0042834), PGN catabolism (GO:0009253) also present		catalytic PGRPs

Comparative Sequence Provenance (Iteration 2 — executed)

Global Needleman–Wunsch % identity of F5HMW5 vs a PGRP panel (identity over aligned columns):

Reference	Class	% identity to F5HMW5
Human PGLYRP2 (Q96PD5)	catalytic amidase	57.8
Drosophila PGRP-LB (Q8INK6)	catalytic amidase	56.7
Drosophila PGRP-LC (Q9GNK5)	recognition receptor	47.8
Drosophila PGRP-SC1 (Q9VXN9)	catalytic amidase	44.3
Drosophila PGRP-SA (Q9VYX7)	recognition (Lys-type)	36.7

Diagnostic catalytic-cysteine motif (anchored on the zinc-His GHR, C-terminal catalytic region):

Protein	Class	Motif	Catalytic Cys
F5HMW5 (target)	catalytic	GHRQVRTTECPGDRL (C178)	present
Drosophila PGRP-LB	catalytic	GHRQVRDTECPGGRL	present
Human PGLYRP2	catalytic	GHRQLVRTDCPGDALF	present
Drosophila PGRP-LC	recognition	motif diverged (no aligned Cys)	absent
Drosophila PGRP-SA	recognition	motif diverged (no aligned Cys)	absent

F5HMW5 retains the full catalytic set: HH doublet (H60/H61), zinc-His **GAH** (H118), catalytic His **GHR** (H171), catalytic Cys **ECPG** (C178), plus groove Tyr (Y56/Y63/Y69). *Caveat:* the strict **GHR** anchor also failed for catalytic PGRP-SC1 (divergent motif window), so motif-absence alone is not proof of non-catalysis for every paralog; the catalytic-vs-recognition contrast rests on the aligned catalytic Cys, consistent with published PGRP biochemistry.

GO Curation Implications (leads — require curator verification)

GO ID	Term	Aspect	Current evidence	Recommended action
GO:0008745	N-acetylmuramoyl-L-alanine amidase activity	MF	IEA+IBA	Retain — core MF (intact catalytic residues + ortholog assays)
GO:0008270	zinc ion binding	MF	IEA	Retain (metalloamidase; supported by active site)
GO:0042834	peptidoglycan binding	MF	IEA	Retain as substrate binding (preferred over "receptor activity")
GO:0016019	peptidoglycan receptor activity	MF	IBA only	Remove / do-not-assert (NOT) — over-annotated via mixed PANTHER node; receptor role belongs to PGRP-LC
GO:0009253	peptidoglycan catabolic process	BP	IEA	Retain — core BP
GO:0045824 (candidate)	negative regulation of innate immune response	BP	not present	Add (lead) — species-matched phenotype (Gendrin 2017; Paredes 2011)
GO:0050830	defense response to Gram-positive bacterium	BP	IBA only	Remove / non-core — wrong chemotype (DAP/Gram-negative) and wrong sign (suppressor)
GO:0005576	extracellular region	CC	IBA	Retain as orthology lead , flag N-terminal signal/TM ambiguity
GO:0016020	membrane	CC	(keyword only)	Do not assert without experimental support (prediction artifact risk)

Avoid "protein binding" as an endpoint; the informative MF is **amidase activity** with **peptidoglycan binding** as substrate recognition.

Mechanistic Scope

Immediate molecular function tested: enzymatic **N-acetylmuramoyl-L-alanine amidase** cleavage of the lactyl-amide bond in peptidoglycan (zinc-dependent), depleting DAP-type muropeptides. Downstream/consequential (not the direct MF): reduced Imd/PGRP-LC signaling, immune tolerance to gut microbiota, and increased *Plasmodium* permissiveness. The seed's "receptor activity" and "Gram-positive defense" are downstream/pathway framings that do not correspond to the gene product's direct biochemical activity; "receptor activity" is additionally a paralog-borrowed function.

Conflicts and Alternatives

- **Paralog confusion:** GO:0016019 and part of GO:0050830 originate from an IBA node combining recognition PGRPs (PGRP-LC/SA-type, mammalian PGLYRP1/3/4) with catalytic ones; F5HMW5 is catalytic. Receptor activity belongs to PGRP-LC.
- **Sign reversal:** Loss-of-function *increases* immunity/parasite clearance, so LB is protective *of the microbe/parasite context*, not a defense effector.
- **Chemotype vs Gram class:** DAP-type PGN occurs in Gram-negative bacteria *and* *Bacillus* (Gram-positive); "Gram-positive defense" is not equivalent to Lys-type specificity and is not the tested chemotype.
- **Isoform/topology:** dual signal/TM prediction implies possible membrane vs secreted isoforms; extracellular pool is inferred, not shown for F5HMW5.

Knowledge Gaps

1. **Direct enzymatic assay on F5HMW5** — checked: none found; matters because amidase assignment currently rests on sequence + orthology. Resolve with recombinant amidase activity on DAP- vs Lys-type PGN.
2. **Substrate specificity of F5HMW5** — checked ortholog (DAP-type); matters for GO:0050830. Resolve by muropeptide-resolved activity assay.

3. **N-terminal processing / localization of F5HMW5** — checked: UniProt/Ensembl show a single transcript (AGAP001212-RA), PE3 (homology-inferred), and a prediction-only (SAM) TM helix; no experimental localization exists. Matters because GO:0005576 vs membrane cannot be decided from current data. Resolve by SignalP/DeepTMHMM analysis plus an N-terminomics/secretion (surface vs supernatant) assay on *A. gambiae* PGRPLB.
4. **Catalytic-dead vs wild-type rescue in Anopheles** — matters to prove the negative-regulation is amidase-dependent.

Discriminating Tests

- Recombinant F5HMW5 amidase assay on purified DAP-type vs Lys-type PGN (distinguishes amidase vs "receptor"; chemotype specificity).
- Catalytic-Cys (C178A) mutant: loss of amidase and loss of Imd down-regulation would confirm mechanism.
- Secretion/topology: signal-peptide cleavage / surface-vs-supernatant fractionation across isoforms.
- Epistasis with PGRPLC knockdown to confirm LB acts by depleting the PGRP-LC ligand.

Curation Leads (verify)

- **Remove/NOT** GO:0016019 (peptidoglycan receptor activity) on F5HMW5 — IBA over-annotation; cite

P 28494453

(PGRPLC is the receptor) and

P 16618604

(LB is a catalytic amidase).

- **Remove/downgrade** GO:0050830 (defense response to Gram-positive bacterium); consider **add** GO:0045824 (negative regulation of innate immune response) with

P 28494453

22118526.

- **Retain** GO:0008745, GO:0008270, GO:0042834, GO:0009253 as core.

- **Retain** GO:0005576 as orthology lead; annotate the N-terminal signal/TM ambiguity; do not assert integral-membrane CC.
- Candidate references with snippets to verify:

P 16618604

"PGRP-LB is a secreted protein"; "amidase that specifically degrades gram-negative bacteria peptidoglycan";

P 28494453

"PGRPLB promotes mosquito permissiveness to *P. falciparum*".

Limitations

Direct biochemical/localization data are for the *Drosophila* ortholog; *A. gambiae* in-vivo evidence (Gendrin 2017) is genetic/RNAi. No F5HMW5-specific enzyme or localization assay was located. Sequence/active-site analysis is computational (public UniProt/PANTHER/Pfam). No fabricated results.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery